

Adoption of Compliant Authoring Tools in Doctoral Education: Three Studies

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Declaration

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Rationale

This thesis is submitted in Journal Format. Include here a short explanation of the rationale for using this format and an account of how the thesis has been constructed, as required by the Presentation of Theses Policy section 13.10.

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Chapter 1

Preface

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Chapter 2

Abstract

This thesis examines the adoption of policy-compliant authoring tools in doctoral education through three constituent papers, submitted in **journal format** per Presentation of Theses Policy section 13. The papers address distinct but complementary aspects of the central question: a cross-institutional survey of tool adoption (Paper 1), an experimental study of compliance enforcement on writing quality (Paper 2), and a qualitative analysis of supervisor perspectives (Paper 3).

Together, the papers establish that policy-compliant tooling is associated with measurable improvements in submission quality, that this association is mediated by supervisor engagement, and that adoption patterns vary systematically with disciplinary culture. The discussion synthesises across the three papers and offers practical recommendations.

All findings are reproducible from the underlying data and code, made available alongside the submitted PDF.

Chapter 3

Acknowledgements

I would like to thank my supervisors for their guidance throughout this research, my examiners for their constructive feedback, and my co-authors for their collaboration on the constituent papers presented in this thesis.

I am grateful to the Alliance Manchester Business School Doctoral Programmes office and the Humanities Doctoral Academy for the administrative and financial support that made this work possible.

Finally, my thanks go to my family and friends, whose patience and encouragement sustained me throughout the writing of this thesis.

Part I

Main body

Chapter 4

Introduction

This thesis is submitted in **journal format** under the University of Manchester *Presentation of Theses Policy* section 13. The format permits doctoral candidates to organise the body of the thesis around a series of self-contained papers, framed by an overarching introduction and discussion.

4.1 Rationale for journal format

The decision to submit in journal format reflects three considerations:

1. **Disciplinary norms.** Research in management and organisational behaviour is typically disseminated through peer-reviewed journal articles rather than monographs. Submitting in journal format aligns the thesis with the conventions of the field.
2. **Empirical structure.** The empirical work falls naturally into three distinct studies, each with its own data, methods, and findings. A standard-format integration would have required compressing or fragmenting the studies in awkward ways.
3. **Career development.** Drafting the constituent papers during the doctoral period accelerates the candidate's transition to academic publishing.

This format choice was made in consultation with the supervisory team during Year 2 of the programme and recorded on the Notice of Submission form.

4.2 Overarching research question

How does the adoption of policy-compliant authoring tools influence the quality and integrity of doctoral thesis submissions?

The question is decomposed into three sub-questions, each addressed by one of the papers:

- **Paper 1** (Chapter 3): *What patterns of adoption do we observe across UK business schools?*
- **Paper 2** (Chapter 4): *Does compliance enforcement causally improve writing quality?*
- **Paper 3** (Chapter 5): *How do supervisors perceive and interpret the role of tooling in supervision?*

4.3 Coherence across the papers

Although each paper stands alone, the three together form a coherent body of work. Paper 1 establishes the empirical landscape of tool adoption; Paper 2 isolates the causal effect of one specific intervention (compliance enforcement); Paper 3 interprets the quantitative findings through the lived experience of supervisors. The General Discussion (Chapter 6) synthesises across the three.

4.4 Statement on the candidate's contribution

The candidate is the first author of all three papers. A detailed contribution declaration is provided at the head of each chapter using the `{contribution="..."}` heading attribute, as required by policy section 13.3. Co-authors include the supervisory team and, for Paper 3, a second qualitative coder.

Chapter 5

Paper 1: Adoption Patterns of Compliant Authoring Tools in UK Business Schools

Contribution statement

I designed the study, drafted and piloted the survey instrument, recruited the participating institutions, collected and cleaned the data, performed the analysis, and wrote the manuscript. Co-author A advised on sampling strategy and reviewed successive drafts.

5.1 Abstract

Cross-sectional survey of doctoral programme directors at 28 UK business schools investigating patterns of adoption of policy-compliant authoring tools. We find substantial heterogeneity across institutions, with adoption rates ranging from 8% to 71%. Adoption is positively associated with institutional size and with the presence of a dedicated research-methods training programme.

5.2 Introduction

Policy-compliant authoring tools have proliferated rapidly over the past five years, but their uptake in doctoral education is poorly documented. This paper provides the first systematic survey of adoption across UK business schools and identifies institutional factors associated with higher uptake.

5.3 Methods

A web-administered survey was sent to doctoral programme directors at all 28 AACSB-accredited UK business schools. The survey instrument was piloted with three programmes and refined based on cognitive interviews. The response rate was 89% ($n = 25$ usable responses).

The dependent variable is the percentage of currently registered PhD candidates using a policy-compliant authoring tool (self-reported by programme directors).

5.4 Results

Adoption rates ranged from 8% to 71% with a mean of 34%. Two institutional factors are associated with higher adoption:

- **Institutional size:** a one-standard-deviation increase in cohort size is associated with a 6.2 percentage-point increase in adoption ($p < 0.05$).
- **Dedicated methods training:** programmes with a formal research-methods sequence show 11.4 percentage-point higher adoption ($p < 0.01$).

5.5 Discussion

The findings suggest that adoption is structurally enabled by institutional resources rather than driven by individual candidate preferences. This has implications for how universities can foster broader uptake.

5.6 Limitations

The cross-sectional design precludes causal claims. Self-report by programme directors may overstate adoption rates if directors are unaware of which tools individual candidates use.

Chapter 6

Paper 2: The Effect of Compliance

Enforcement on Writing Quality: An A/B Experiment

Contribution statement

I designed the experimental protocol, secured ethical approval, recruited participants, randomised treatment assignment, collected outcome data, performed the statistical analysis, and wrote the manuscript. Co-author A advised on the experimental design and Co-author B served as a second blind rater of the outcome variable.

6.1 Abstract

We report a pre-registered randomised experiment in which 120 doctoral candidates were assigned to write a 5,000-word chapter using either a compliance-enforcing authoring tool (treatment) or an unconstrained alternative (control). Compliance enforcement caused a 0.31 standard-deviation improvement in the blinded quality rating (95% CI: 0.09–0.52, $p = 0.006$).

6.2 Introduction

Paper 1 established that adoption of compliant authoring tools varies systematically across institutions. This paper asks whether adoption *causes* improvements in writing quality, or whether the association is confounded by selection. We address the question through a randomised experiment.

6.3 Methods

6.3.1 Participants and randomisation

Candidates ($n = 120$) were recruited from three participating universities and randomised at the individual level to treatment ($n = 60$) or control ($n = 60$). Randomisation was stratified by programme year and institution.

6.3.2 Intervention

The treatment group used a policy-compliance-enforcing authoring tool with structural and stylistic constraints. The control group used an unconstrained alternative.

6.3.3 Outcome measure

The primary outcome is a blinded expert rating of overall writing quality on a 1–7 scale, averaged across two independent raters. Inter-rater reliability was high ($\kappa = 0.78$).

6.4 Results

Treatment caused a statistically significant improvement in the quality rating. Mean quality in the treatment group was 4.92 (SD 0.91); mean in control was 4.61 (SD 0.86). The intent-to-treat effect was 0.31 standard deviations.

6.5 Discussion

The experimental design provides causal identification of the effect of compliance enforcement on writing quality. The magnitude of the effect is comparable to other educational interventions of similar intensity.

6.6 Limitations

The 5,000-word chapter task may not generalise to the full thesis-writing process. Longer-term effects on submission outcomes are not measured here.

Chapter 7

Paper 3: Supervisor Perspectives on Authoring Tooling in Doctoral Supervision: A Qualitative Study

Contribution statement

I designed the interview protocol, recruited and interviewed all 15 supervisor participants, transcribed the interviews, conducted the primary thematic analysis, and wrote the manuscript. Co-author C served as a second independent coder for inter-rater reliability and reviewed the manuscript.

7.1 Abstract

Semi-structured interviews with 15 doctoral supervisors at three UK business schools examined how supervisors interpret and respond to candidate use of policy-compliant authoring tools. Three themes emerged: *delegation of compliance work*, *displacement of intellectual conversation*, and *re-distribution of authority over thesis structure*.

7.2 Introduction

Papers 1 and 2 establish that policy-compliant tooling is increasingly adopted and that adoption causally improves writing quality. But the quantitative work cannot illuminate the *meaning* of these tools for the supervisory relationship. This paper addresses that gap through qualitative interviews.

7.3 Methods

7.3.1 Sampling

A purposive sample of 15 supervisors was recruited from three universities, balanced by gender, career stage, and disciplinary sub-field. Sampling continued until thematic saturation was reached, following Brown and Davies (2023).

7.3.2 Data collection

Semi-structured interviews of 45–75 minutes were conducted via video conference and audio-recorded with consent.

7.3.3 Analysis

Transcripts were coded using reflexive thematic analysis. The candidate and Co-author C independently coded the first five transcripts; coding inconsistencies were resolved through discussion, after which the candidate completed the remaining coding.

7.4 Findings

7.4.1 Theme: delegation of compliance work

Supervisors framed compliance tooling as a “delegation” that freed the supervisor-candidate conversation from procedural concerns:

Before, I would spend half of our meeting going over the declaration template and the copyright statement. Now the tool does that. We can actually talk about the argument.

— Supervisor 7, senior professor

7.4.2 Theme: displacement of intellectual conversation

A minority of supervisors expressed concern that the tools could *displace* intellectual conversation by making structural choices invisible:

The candidate doesn’t even see the decisions the tool is making. There’s a risk that they internalise these conventions without ever interrogating them.

7.4.3 Theme: re-distribution of authority

The most consistent theme was a perceived shift in authority over thesis structure from the supervisor to the institution (via the tool).

7.5 Discussion

The findings complement the quantitative results in Papers 1 and 2 by showing that the *meaning* of tooling adoption is contested. The implication for institutional policy is that tool deployment should be accompanied by supervisor training that addresses these tensions directly.

7.6 Limitations

The sample is restricted to three universities and to business and management disciplines.

Chapter 8

General Discussion

This chapter synthesises across the three papers.

8.1 Synthesis of findings

The three papers together establish a coherent picture:

- **Paper 1** documents substantial cross-institutional variation in adoption and identifies institutional size and methods training as enablers.
- **Paper 2** provides causal evidence that compliance enforcement improves writing quality at the chapter level.
- **Paper 3** illuminates the meanings supervisors attach to tooling and identifies tensions around delegation, displacement, and authority.

A common thread is that tooling is most effective when embedded in an institutional infrastructure that includes supervisor engagement. Pure technical deployment is insufficient.

8.2 Theoretical contribution

The thesis advances the *adoption-as-infrastructure* perspective by treating policy-compliant authoring tools as components of a broader institutional ecology rather than as discrete artifacts adopted by individuals.

8.3 Practical implications

Three practical recommendations follow:

1. **Institutional:** methods-training programmes should integrate authoring-tool training as a first-class component.
2. **Supervisory:** supervisor development should explicitly address the tensions identified in Paper 3.
3. **Tool design:** compliance tools should expose, rather than hide, the structural decisions they enforce, mitigating the displacement concern.

8.4 Limitations across the papers

The three papers share a UK business-school context. Replication in other national systems and disciplinary cultures is a priority for future work.

8.5 Future research agenda

Three directions stand out:

- A multi-country comparative study.
- A longitudinal panel that tracks candidates from registration to submission.
- A design experiment that tests alternative tool affordances against the supervisor concerns surfaced in Paper 3.

8.6 Closing remarks

The findings reinforce an argument from The University of Manchester (2026): the *infrastructure* of academic writing shapes the integrity of research outputs. As tooling continues to evolve, the boundary between writing and computation will continue to blur.

Appendix A

Appendix A: Supplementary Material

A.1 A.1 Survey instrument (Paper 1)

The full survey instrument used in Paper 1 is reproduced below. Items were piloted with three programme directors and refined based on cognitive interviews.

Survey administration

The survey was administered online via Qualtrics between September and November 2025. Programme directors received an initial invitation followed by two reminder emails at two-week intervals.

A.2 A.2 Experimental protocol (Paper 2)

The pre-registered protocol for the Paper 2 experiment is available at the project's Open Science Framework registry.

A.3 A.3 Interview topic guide (Paper 3)

The semi-structured interview topic guide developed for Paper 3 is reproduced below.

1. **Opening:** Tell me about your current doctoral supervisees.
2. **Tooling:** Are any of your supervisees using a policy-compliant authoring tool? Which tool? How did they come to use it?
3. **Experience:** What has been your experience of working with candidates who use these tools?

4. **Tensions:** Have there been moments of difficulty or tension related to the tool?
5. **Authority:** Whose decisions does the tool encode? Do you see this as appropriate?
6. **Future:** Would you encourage other candidates to use the tool? Under what circumstances?

Reproducibility

All quantitative results in Papers 1 and 2 are reproducible from the analysis code at the project repository. Qualitative coding for Paper 3 is documented in a separate codebook.

- [] Brown, S.K. and Davies, E.R. (2023) ‘Qualitative methods in doctoral supervision research’, *Studies in Higher Education*, 48(7), pp. 1102–1119.
- [] The University of Manchester (2026) ‘Presentation of Theses Policy, version 12’. Available at: <https://documents.manchester.ac.uk/display.aspx?DocID=7420>.